

Customer Churn Analysis Report

1. Project Title

Customer Churn Analysis Dashboard

2. Objective

The objective of this project is to analyse customer churn using telecom customer data and identify the major factors that influence customer retention and customer loss. The analysis focuses on churn patterns, contract type, internet service, payment method, monthly charges, and customer tenure. The final dashboard helps understand why customers are leaving and provides useful recommendations to improve customer retention.

3. Dataset Description

The dataset used in this project is a telecom customer churn dataset. It contains customer-related details such as customer ID, gender, senior citizen status, partner and dependent information, tenure, phone service, internet service, online services, contract type, payment method, monthly charges, total charges, and churn status.

The churn column indicates whether a customer has left the company or continued with the service.

4. Tools Used

Power BI was used for creating the dashboard and visual analysis. The dataset was loaded, cleaned, and analysed using Power BI visuals and measures. Charts and KPI cards were used to present churn trends and customer behaviour clearly.

5. Key Performance Indicators

The dashboard includes the following key metrics:

Total Customers: 7,043

Churned Customers: 1,869

Retained Customers: 5,174

Churn Rate: 26.54%

These KPIs give a quick overview of the overall customer churn situation.

6. Dashboard Visuals

The dashboard includes the following visuals:

Churn Distribution: Shows the percentage of churned and retained customers.

Churn by Contract Type: Shows churn based on month-to-month, one-year, and two-year contracts.

Churn by Internet Service: Shows churn based on internet service type such as fiber optic, DSL, and no internet service.

Churn by Payment Method: Shows churn based on different payment methods.

Average Monthly Charges by Churn: Compares the average monthly charges of churned and retained customers.

Average Tenure by Churn: Compares the average tenure of churned and retained customers.

7. Key Insights

Out of 7,043 customers, 1,869 customers have churned, giving an overall churn rate of 26.54%.

Month-to-month contract customers have the highest churn compared to one-year and two-year contract customers. This shows that customers with short-term contracts are more likely to leave.

Fiber optic internet service customers show higher churn compared to DSL and customers with no internet service. This may indicate possible issues related to pricing, service quality, or customer expectations.

Customers using electronic check as the payment method have higher churn compared to other payment methods.

Churned customers have higher average monthly charges compared to retained customers. This suggests that higher charges may be one of the reasons for customer dissatisfaction.

Customers with lower tenure are more likely to churn, while long-term customers are more likely to stay with the company.

8. Recommendations

The company should encourage month-to-month contract customers to shift to one-year or two-year contracts by providing discounts or special offers.

The company should focus on improving service quality and customer support for fiber optic internet users.

Customers with high monthly charges should be given personalized offers, discounts, or better plan options to reduce churn.

The company should promote automatic payment methods instead of electronic checks to improve customer retention.

New customers should be given extra attention during the first few months through onboarding support, follow-up communication, and retention offers.

Customers with low tenure should be monitored carefully because they are more likely to churn.

9. Conclusion

This project helped analyse customer churn patterns and identify the major factors affecting customer retention. The Power BI dashboard clearly shows that contract type, internet service, payment method, monthly charges, and tenure have a strong relationship with churn. By focusing on high-risk customer groups and applying suitable retention strategies, the company

can reduce customer loss and improve long-term customer satisfaction.



